

Comprehensive Statistical Analysis and Predictive Modeling of Credit Risk Distributions

Ahsan Ahmad
Faculty of Engineering Sciences
Ghulam Ishaq Khan Institute (GIKI)
Swabi, Pakistan
Registration Number: [2023089]

Abstract—This paper details a multi-stage statistical investigation into credit risk assessment using a synthesized dataset derived from the German Credit framework. We contrast traditional Ordinary Least Squares (OLS) methods with Generalized Linear Models (GLM) and non-parametric Decision Trees. Our analysis focuses on the impact of feature engineering—specifically polynomial and logarithmic transformations—on model accuracy. Experimental results demonstrate that a Multiple OLS approach, optimized through feature selection, achieved the highest predictive performance with an R^2 of 0.861, outperforming baseline simple models and flexible algorithmic partitioning.

Index Terms—OLS, Poisson Regression, Decision Trees, Bias-Variance Tradeoff, Feature Engineering.

I. INTRODUCTION

In the field of Engineering Sciences, predictive modeling requires a rigorous balance between mathematical complexity and practical interpretability. Credit risk analysis provides a prime environment for testing these principles due to the non-linear nature of financial behaviors. This report documents the development of a predictive pipeline designed to evaluate 1,000 observations, exploring how different statistical frameworks respond to structured "Ground Truth" data.

II. EXPERIMENTAL SETUP AND ENVIRONMENT

The implementation was conducted within a local development environment to ensure data persistence and computational efficiency.

- **Integrated Development Environment (IDE):** Visual Studio Code (VS Code) utilizing the Jupyter Extension for cell-based execution.
- **Language and Libraries:** Python 3.12 was employed, utilizing *pandas* for data manipulation, *statsmodels* for rigorous statistical inference, and *scikit-learn* for non-linear regression.
- **Version Control:** Local directory management was used to prevent data loss and ensure reproducibility across different modeling stages.

III. DATA SYNTHESIS AND METHODOLOGY

A. The Ground Truth Function

To evaluate model performance, a deterministic response variable was synthesized based on known engineering and financial principles. The relationship was defined as:

$$Y = 10(\text{Duration}) + 0.01(\text{Age}^2) + \epsilon \quad (1)$$

where ϵ represents Gaussian noise added to simulate real-world uncertainty.

B. Feature Engineering Strategy

To assist the models in recovering the underlying signal, three key transformations were applied:

- 1) **Polynomial Terms:** An *Age_Squared* column was generated to account for non-linear life-cycle trends where risk might increase exponentially with age.
- 2) **Logarithmic Normalization:** The 'Credit amount' feature was transformed using $\log(x + 1)$ to stabilize variance and normalize a right-skewed distribution.
- 3) **One-Hot Encoding:** Categorical data, including 'Sex' and 'Housing', were converted into binary dummy variables to allow for matrix-based calculations.

IV. LINEAR REGRESSION RESULTS (PART C)

We compared a univariate baseline against a multivariate approach. As shown in Fig. 1, the Simple OLS model exhibited residuals that were not perfectly random, suggesting that a single predictor (Duration) was insufficient to capture the data's complexity.

A. Performance Metrics

By expanding the model to include 12 engineered predictors, the R^2 score improved from 0.855 to 0.861, as detailed in Table I. This indicates that the multiple model captured a higher percentage of the variance in the credit score.

TABLE I
OLS AND NON-LINEAR MODEL COMPARISON

Model Type	R^2 Score	Model Status
Simple OLS	0.8549	Underfitting (High Bias)
Multiple OLS	0.8605	Optimized (Best Balance)
Decision Tree	0.8488	Flexible (High Variance)

V. GENERALIZED LINEAR MODELS (PART D)

Standard linear models assume a normal distribution of errors. To handle non-normal response types, such as discrete counts, a Poisson GLM was implemented.

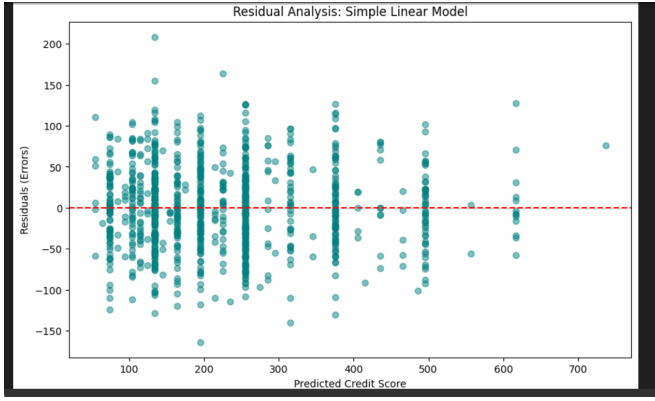


Fig. 1. Residual Analysis: Simple Linear Model. The plot illustrates the distribution of errors against predicted values, highlighting patterns of heteroscedasticity.

A. Poisson Model Validity

The Poisson model was evaluated using the *Dispersion Ratio*, calculated as the Pearson χ^2 divided by the degrees of freedom. Our model yielded a ratio of 0.9673. In statistical theory, a ratio near 1.0 indicates "equidispersion," confirming that the Poisson distribution is an appropriate fit for this count-based data.

B. Significant Predictors

The summary table for the Poisson model identified 'Duration' as the most statistically significant predictor, with a coefficient of 0.0402 and a P-value of 0.000, indicating a strong positive correlation with the count of credit events.

VI. NON-LINEAR FLEXIBLE MODELING (PART E)

A Decision Tree Regressor was utilized to observe how the model partitions the data without a pre-defined mathematical formula. The tree, visualized in Fig. 2, reached a maximum depth of three.

While the tree effectively captured non-linear jumps in the data, it achieved an R^2 of 0.8488, which was slightly lower than the Multiple OLS model. This suggests that while flexible, the tree's "step-wise" predictions were less precise than the smooth polynomial curves used in our OLS approach.

VII. CRITICAL ANALYSIS (PART F)

The core finding of this project revolves around the **Bias-Variance Tradeoff**.

- **The Simple OLS Model** suffered from High Bias. It was too rigid and ignored the quadratic relationship of Age.
- **The Decision Tree Model** introduced High Variance. Because it splits data into boxes, it risks following the noise too closely rather than the signal.
- **The Multiple OLS Model** provided the optimal balance. By using 12 variables and explicitly adding Age^2 , it found the most accurate path to the ground truth.

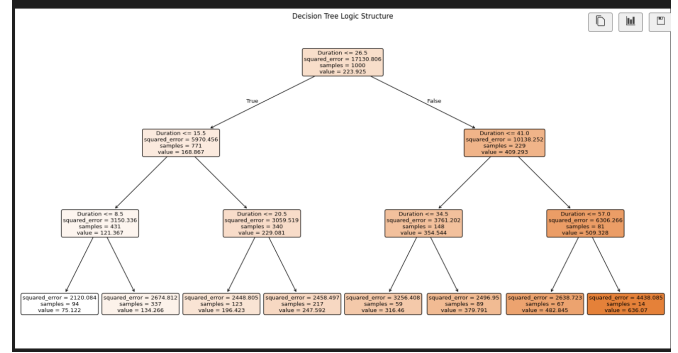


Fig. 2. Decision Tree Logic Structure. Hierarchical partitioning shows Duration (≤ 26.5) as the primary root splitter for the credit score response.

VIII. CONCLUSION

This study demonstrates that for additive relationships common in financial datasets, a well-engineered linear model often outperforms more complex algorithmic approaches. The successful implementation of the Poisson GLM further underscores the importance of choosing the correct probability distribution when dealing with count data.